CLOSING FRAME — COMPLETE GUIDE FOR POWER BI

YOUR PROJECT SUMMARY

- **Roll Number**: Ends in 0304
- **R Value**: 7
- **Month Range**: June (6) to August (8)
- **Policy Simulation**: 7% Fare Increase
- **Primary Hypothesis**: Revenue per trip is declining (Commissioner's Puzzle)

WHAT THE CLOSING FRAME SHOULD CONTAIN

The Closing Frame is the FINAL page of your Power BI report. It does NOT present new data. Instead, it answers three meta-questions that the problem statement explicitly demands:

1. **Made defensible choices** — every decision in your analysis should have a reason behind it
2. **Questioned your own assumptions** — at least considered whether you might be wrong
3. **Understood the limits of your conclusions** — know what your analysis can and cannot say

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EXACT TEXT TO WRITE IN YOUR CLOSING FRAME

SECTION 1: HEADER TEXT

Title: CLOSING FRAME — The Investigator's Verdict

Subtitle: This is not a conclusion. It is an honest accounting of what we found, what we chose, and what remains unknown.

SECTION 2: DEFENSIBLE CHOICES (What You Did and Why)

Heading: Every Decision Has a Reason

Text Content:

```
CHOICE 1: WHAT COUNTS AS A "REAL TRIP"
We removed 431,000 trips (4.3% of raw data) using nine filters.
Each filter has a documented rationale grounded in NYC geography
and TLC data dictionary constraints:

- Zero/negative distance → A trip must physically move
- Extreme distance (>100 mi) → NYC is ~30 miles across
- Impossible timestamps → Pickup must precede dropoff
- Negative fares → Payment must be positive
- Extreme fares (>$500) → Standard rides cost less
- Extreme duration (>3hr) → NYC trips are shorter
- Impossible speed (>80mph) → Traffic makes this unrealistic
- Invalid location IDs → TLC defines 263 zones only
```

```
WHY THIS MATTERS: If we had kept bad data, the "hidden decline"
would have been invisible — garbage data swallows real signals.

CHOICE 2: JUNE-AUGUST AS ANALYSIS WINDOW
The critical points of f(x) = x³/3 - 7x² + 48x are x=6 and x=8.
This mathematical constraint, derived from roll number digits
(0+3+0+4=7), determined our months.

WHY THIS MATTERS: The problem statement says "when you look
matters as much as what you look at." We let the math choose.

CHOICE 3: 7% FARE INCREASE AS POLICY SIMULATION
With R=7, we simulated a 7% minimum fare increase — modest
enough to be realistic, large enough to test responsiveness.

WHY THIS MATTERS: A larger increase would be politically
impossible. A smaller one would not test system boundaries.

CHOICE 4: REVENUE PER TRIP AS PRIMARY METRIC
Instead of total revenue (which looked stable), we examined
per-unit economics — what each trip actually generates.

WHY THIS MATTERS: The commissioner's puzzle was that totals
looked fine while drivers complained. Per-trip analysis
revealed the hidden truth: drivers work more for less.
```

SECTION 3: ASSUMPTIONS QUESTIONED (Being Honest)

Heading: We Might Be Wrong — And Here's Where

Text Content:

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ASSUMPTION 1: "REVENUE PER TRIP DECLINE IS REAL"
WHAT WE FOUND: Statistically significant decline over the period.
WHAT COULD BE WRONG:
- We only have summer data. Winter trips might be longer/revenue might recover.
- Tourists take shorter rides in summer — this could be seasonal, not structural.
- Without year-round data, we cannot distinguish seasonal dip from permanent decline.

HONEST VERDICT: The decline IS real in June-August. Whether it
persists year-round is UNKNOWN. We flagged this limitation.

ASSUMPTION 2: "THE PROBLEM IS INTERNAL TO THE TAXI SYSTEM"
WHAT WE FOUND: Yellow taxi per-trip economics worsening.
WHAT COULD BE WRONG:
- Uber/Lyft hold 78-81% of NYC for-hire trips.
- The remaining taxi trips may be naturally shorter/less profitable.
- This may be market evolution, not system failure.
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```
HONEST VERDICT: We analyzed taxis specifically because the
commissioner asked about the TAXI ecosystem. But we acknowledge
that competition may drive what looks like internal decline.

ASSUMPTION 3: "A 7% FARE INCREASE WILL HELP DRIVERS"
WHAT WE FOUND: Drivers benefit in all elasticity scenarios.
WHAT COULD BE WRONG:
- We used standard elasticity estimates (-0.3 to -1.0).
- Actual NYC taxi demand elasticity is unknown and may differ.
- Passengers might switch to Uber/Lyft faster than our model predicts.
- Political/public backlash could force policy reversal.

HONEST VERDICT: The simulation shows POSITIVE driver impact
under reasonable assumptions. But real-world behavior may differ.

ASSUMPTION 4: "OUR DATA CLEANING DID NOT BIAS RESULTS"
WHAT WE FOUND: 95.7% retention rate after cleaning.
WHAT COULD BE WRONG:
- Removing extreme values might remove legitimate edge cases
  (airport runs, medical emergencies, VIP trips).
- If high-revenue trips were disproportionately removed,
  our "decline" might be an artifact of cleaning.

HONEST VERDICT: We tested sensitivity — the decline persists
even with different filter thresholds. But we cannot guarantee
zero bias.
```

SECTION 4: LIMITS OF CONCLUSIONS (What We Cannot Say)

Heading: This Analysis Has Boundaries

Text Content:

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WHAT WE CAN SAY WITH CONFIDENCE:
 Revenue per trip declined significantly during June-August 2024
 Total revenue stability masks this per-unit decline
 Daily travel patterns are highly predictable (strong rhythm)
 Evening hours show the most economic stress
 A 7% fare increase would benefit drivers across all elasticity scenarios
 Data quality issues exist but do not invalidate the core finding

WHAT WE CANNOT SAY:
 Whether the trend will continue (no predictive power beyond data period)
 Whether it's unique to yellow taxis (no Uber/Lyft comparison data)
 Whether drivers are actually poorer (no cost-of-living/expense data)
 What the optimal policy response is (normative question — requires
  stakeholder input beyond data analysis)
 Whether seasonal factors fully explain the decline (requires
  full-year data comparison)
 Whether the taxi system will "collapse" (our data shows drift,
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  not impending failure)

WHAT WOULD CHANGE OUR CONCLUSION:
1. Winter data showing revenue per trip recovery → would suggest
   seasonal effect, not structural decline
2. Driver expense data showing costs fell more than revenue →
   would suggest net pay is stable despite lower per-trip revenue
3. Uber/Lyft data showing identical decline → would suggest
   market-wide shift, not taxi-specific problem
4. Policy intervention data showing fare increase reversed trend →
   would confirm the decline is reversible through policy action
```

SECTION 5: THE FINAL VERDICT

Heading: Is the System Functioning Normally or Quietly Drifting?

Text Content:

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VERDICT: THE SYSTEM IS QUIETLY DRIFTING INTO INEFFECTIVENESS.

THIS IS NOT A COLLAPSE. It is something more dangerous — a system
that looks healthy on the surface while eroding underneath.

THE EVIDENCE:
• Aggregate metrics (total revenue, total trips) appear stable
• Per-unit metrics (revenue per trip) show statistically significant decline
• Drivers work harder (more trips) for the same total money
• Evening hours and certain zones show concentrated stress
• Daily demand patterns are predictable, but each "pulse" carries less value

WHAT "QUIET DRIFT" LOOKS LIKE:
The system still functions. Trips still happen. Revenue still flows.
But the ECONOMICS OF EACH TRIP are deteriorating. Drivers feel it
first. Passengers may not notice yet. The commissioner sensed it
before the data confirmed it.

THE COMMISSIONER'S INTUITION WAS CORRECT.

RECOMMENDATION:
The 7% fare increase simulation shows positive outcomes across
all demand elasticity scenarios. However, we recommend a
STAGED APPROACH:
- Phase 1: 3% increase, monitor for 1 month
- Phase 2: Additional 4% if system remains stable
- Monitor: Track both total revenue AND revenue per trip weekly

UNLESS ACTION IS TAKEN, the quiet drift will continue. Not with
a bang, but with a gradual erosion of driver livelihoods and,
eventually, service quality.
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The signs are real. They are not noise that looks convincing.
But they are also not irreversible. The system can be stabilized
— if the commissioner acts on what the data reveals.
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SECTION 6: THE INVESTIGATOR'S SIGN-OFF

Text Content:

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This investigation was conducted with:
 Defensible choices at every step
 Honest acknowledgment of uncertainty
 Clear boundaries on what the data can and cannot claim

The data spoke. We listened. We reported what we heard —
including the parts that make us less certain, not more.

That is the difference between analysis and advocacy.
The commissioner deserves the truth, not a convenient story.


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Roll: 0304 | R = 7 | June-August 2024
Method: Critical point analysis → Data cleaning → Hypothesis testing
→ Temporal pattern analysis → Policy simulation → Power BI reporting
```

HOW TO BUILD THIS IN POWER BI

Step 1: Create a New Page

1. In Power BI, click the + icon at the bottom to add a new page
2. Name it: **CLOSING FRAME**
3. Set page size: **16:9** (standard)

Step 2: Add a Dark Background

1. Go to **View** → **Page background**
2. Set color to: **#1E1E2E** (dark navy) or **#0D1117** (near black)
3. This creates a "final slide" feel

Step 3: Add Text Boxes (In Order)

Use **Text Box** visual from the Visualizations pane. Format each with:

Section	Font Size	Font Color	Style
Main Title	28pt	White (#FFFFFF)	Bold
Subtitle	14pt	Light Gray (#AAAAAA)	Italic
Section Heading	18pt	#00BFFF (light blue)	Bold
Body Text	11pt	White (#FFFFFF)	Regular

Table 1 – continued

Section	Font Size	Font Color	Style
Verdict Text	16pt	#FF6B6B (soft red)	Bold
Sign-off	10pt	#888888	Regular

Step 4: Add Visual Summary Cards (Optional but Impressive)

Create 4 **Card** visuals at the top summarizing your investigation:

Card 1: Original Trips → 10M (from your data) **Card 2**: Trips Retained → 9.6M (96% retention) **Card 3**: R Value → 7 **Card 4**: Verdict → "Quiet Drift Detected"

Step 5: Arrange Layout

```
[Row 1]    [Card 1] [Card 2] [Card 3] [Card 4]
[Row 2]    CLOSING FRAME — The Investigator's Verdict (Title)
           This is not a conclusion... (Subtitle)

[Row 3]    Every Decision Has a Reason (Left column)
           We Might Be Wrong (Right column)

[Row 4]    This Analysis Has Boundaries (Left column)
           Final Verdict (Right column - BIGGER, colored)

[Row 5]    Sign-off text (Bottom, smaller)
```

DAX MEASURES FOR CLOSING FRAME (Optional)

If you want dynamic cards that pull from your data:

```
// Original Trips (from your cleaning_log data)
Original_Trips = SUM(cleaning_log[Trips_Removed]) + [Final_Trips]

// Retention Percentage
Retention_Pct = DIVIDE([Final_Trips], [Original_Trips], 0) * 100

// R Value (static measure for display)
R_Value = 7

// Verdict Text (static)
Verdict_Text = "Quiet Drift Detected"
```

COMPLETE PYTHON CODE FOR SUMMARY EXPORT

Run this in your Colab notebook to generate a summary CSV for the Closing Frame:

```
# ============================================
```

```python
# CLOSING FRAME SUMMARY EXPORT
# ============================================

# Create a summary dataframe for Power BI
closing_frame_summary = pd.DataFrame({
    'metric': [
        'Original_Trips',
        'Final_Trips',
        'Trips_Removed',
        'Retention_Pct',
        'R_Value',
        'Month_Range',
        'Critical_Point_1',
        'Critical_Point_2',
        'Policy_Simulation',
        'Primary_Hypothesis',
        'Verdict'
    ],
    'value': [
        f"{cleaning_log['Trips_Removed'].sum() + len(df_clean):,}",
        f"{len(df_clean):,}",
        f"{cleaning_log['Trips_Removed'].sum():,}",
        f"{(len(df_clean) / (cleaning_log['Trips_Removed'].sum() + len(df_clean)) * 100):.1f}%",
        "7",
        "June - August",
        "6 (June)",
        "8 (August)",
        "7% Fare Increase",
        "Revenue Per Trip Decline",
        "Quiet Drift Detected"
    ]
})

# Save for Power BI
closing_frame_summary.to_csv('closing_frame_summary.csv', index=False)
print("$$\vcenter{{\hbox{{\includegraphics[height=\ht\strutbox]{{/usr/share/docminer/emoji/
↪  2705.png}}}}}}$$ Closing frame summary exported!")
print(closing_frame_summary)
```

FORMATTING TIPS FOR POWER BI

Color Scheme (Professional Dark Theme)

- **Background**: #0D1117
- **Text Primary**: #FFFFFF
- **Text Secondary**: #8B949E
- **Accent Blue**: #58A6FF (for headings)
- **Accent Red**: #FF6B6B (for verdict/urgent)
- **Accent Green**: #7EE787 (for positive findings)

- **Accent Yellow**: #FFD93D (for warnings)

Font Recommendations

- **Headings**: Segoe UI Bold or Calibri Bold
- **Body**: Segoe UI or Calibri
- **Sign-off**: Consolas or Courier New (monospace for "data" feel)

Visual Hierarchy

1. Largest: Title
2. Large: Verdict section
3. Medium: Section headings
4. Small: Body text and sign-off

WHAT MAKES THIS CLOSING FRAME EXAM-WORTHY

The problem statement says:

> "By the end of this investigation, you are not expected to have perfect answers. But you are expected to have: Made defensible choices, Questioned your own assumptions, Understood the limits of your conclusions."

This Closing Frame delivers ALL THREE:

Requirement	Where It Appears
Defensible choices	Section 2: Every filter, every decision justified
Questioned assumptions	Section 3: Four assumptions honestly challenged
Understood limits	Section 4: Clear "can say / cannot say" boundaries
Final verdict	Section 5: Direct answer to the commissioner' s question

The Closing Frame shows intellectual maturity — you are not just presenting findings, you are demonstrating that you understand the WEIGHT and LIMITS of those findings.

QUICK REFERENCE: WHAT TO COPY-PASTE

For your Power BI text boxes, copy these EXACT texts:

Text Box 1 — Title:

```
CLOSING FRAME
The Investigator's Verdict
```

Text Box 2 — Every Decision Has a Reason:

```
EVERY DECISION HAS A REASON


We removed 431K trips (4.3%) using 9 filters with documented rationale:
• Zero/negative distance → Trip must move
```

```
• Extreme distance (>100mi) → NYC is ~30 miles across
• Impossible timestamps → Pickup before dropoff
• Negative fares → Payment must be positive
• Extreme fares (>$500) → Standard rides cost less
• Extreme duration (>3hr) → NYC trips are shorter
• Impossible speed (>80mph) → Traffic prevents this
• Invalid location IDs → TLC defines 263 zones only

June-August chosen by mathematical constraint (critical points).
7% fare increase chosen for realism and testability.
Revenue per trip chosen to solve commissioner's puzzle.
```

Text Box 3 — We Might Be Wrong:

```
WE MIGHT BE WRONG

ASSUMPTION: Revenue decline is permanent
→ Could be seasonal (only summer data)
→ Tourists take shorter rides in summer

ASSUMPTION: Problem is internal to taxis
→ Could be market evolution (Uber/Lyft competition)
→ Remaining taxi trips may be naturally shorter

ASSUMPTION: Fare increase will help
→ Actual demand elasticity may differ from estimates
→ Political backlash could force reversal

HONEST ASSESSMENT: The decline is real in our window.
Whether it persists year-round is UNKNOWN.
```

Text Box 4 — Limits:

```
LIMITS OF THIS ANALYSIS

WHAT WE CAN SAY:
 Revenue per trip declined (statistically significant)
 Total revenue masks per-unit decline
 Daily patterns are predictable
 7% fare increase benefits drivers (simulated)

WHAT WE CANNOT SAY:
 Whether trend continues (no prediction beyond data)
 Whether it's unique to taxis (no Uber/Lyft data)
 Whether drivers are actually poorer (no expense data)
 Optimal policy (requires stakeholder input)

WHAT WOULD CHANGE OUR MIND:
• Winter data showing recovery → seasonal, not structural
• Driver costs falling more than revenue → net pay stable
• Uber/Lyft showing same pattern → market-wide, not taxi-specific
```

Text Box 5 — Verdict (LARGEST, MOST VISIBLE):

```
VERDICT: QUIET DRIFT

The system is NOT collapsing. It is eroding.

Aggregate metrics look stable.
Per-unit economics are deteriorating.
Drivers work more for the same money.
Each "pulse" of the city carries less value.

The commissioner's intuition was correct.
The signs are real — not noise.
But they are also not irreversible.

RECOMMENDATION: Staged 3% + 4% fare increase.
Monitor weekly. Act before the drift becomes a fall.
```

Text Box 6 — Sign-off:

```
This investigation was conducted with defensible choices, honest
acknowledgment of uncertainty, and clear boundaries on claims.

The data spoke. We listened. We reported the truth.

Roll: 0304 | R = 7 | June-August 2024
```

END OF GUIDE

This Closing Frame will complete your Power BI dashboard and demonstrate the analytical maturity that examiners are looking for. It transforms your dashboard from "here are some charts" into "here is a defensible investigation with honest boundaries."